Hello WORLD

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1. This looks like the definition of a derivative I am going to apply L'hoptials rule:
$$\lim_{x \to \pi} \frac{\cos(x) + \sin(2x) + 1}{x^2 - \pi}$$

2. Let's find the radius of convergence of the power series $\sum_{n=0}^{\infty} (5x-4)^n$ We can apply the ratio test to yeild the following: $\lim_{n \to \infty} |\frac{(5x-4)(5x-4)^n}{(5x-4)^n}|$